

Global Pathogen Analysis Service

User Portal Guide



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Preface

This preface contains the following sections:

- [Documentation accessibility](#)
- [Related Resources](#)
- [Diversity and Inclusion](#)

Documentation accessibility

For information about Oracle's commitment to accessibility, visit the Oracle Accessibility Program website at <http://www.oracle.com/pls/topic/lookup?ctx=acc&id=docacc>.

Related Resources

All documentation and other supporting materials are available on the [Oracle Help Center](#).

Diversity and Inclusion

Oracle is fully committed to diversity and inclusion. Oracle respects and values having a diverse workforce that increases thought leadership and innovation. As part of our initiative to build a more inclusive culture that positively impacts our employees, customers, and partners, we are working to remove insensitive terms from our products and documentation. We are also mindful of the necessity to maintain compatibility with our customers' existing technologies and the need to ensure continuity of service as Oracle's offerings and industry standards evolve. Because of these technical constraints, our effort to remove insensitive terms is ongoing and will take time and external cooperation.

1

Learn more

Spend a few moments familiarizing yourself with the ways the Global Pathogen Analysis Service (GPAS) can help with sequencing the SARS-CoV-2 virus, identifying unique mutations or variants of concern, and more.

- [Global Pathogen Analysis Service](#)
The Global Pathogen Analysis Service (GPAS) is a global, cloud-based system for pathogen sequencing analysis.
- [Global Pathogen Analysis Service Users](#)
The following table describes GPAS user role types.
- [Pipeline components](#)
GPAS supports various pipeline components for the analysis of viral pathogens and quality control outputs.
- [Get support](#)
You can get support for GPAS in one of the following ways:

Global Pathogen Analysis Service

The Global Pathogen Analysis Service (GPAS) is a global, cloud-based system for pathogen sequencing analysis.

To use GPAS, you upload genomic sample data for SARS-CoV-2. You can view consensus sequences and downstream analysis. You can also view relatedness data for cluster analysis and new variant identification. Once the sample data runs through analysis, you can release it to a Shared Data Pool. The data in GPAS is visible in an interactive dashboard, and you can use it for local lab analysis and to make informed public health policy decisions.

GPAS includes the following:

- An upload client that securely sends anonymized sample data to the GPAS cloud (available as a command-line tool and graphical user interface).
- An assembly and analysis pipeline running in the cloud. This includes the Viridian assembly pipeline, Nextclade, PangoLEARN/Scorpio, AIn2type, and other tools. Scalable Pathogen Pipeline Platform ([SP3](#)) and Kubernetes orchestrate these tools.
- Technology that finds near genetic neighbours for a given sequence, called FindNeighbour4 ([FN4](#)).

Global Pathogen Analysis Service Users

The following table describes GPAS user role types.

Table 1-1 GPAS Users

User	Description
Research Scientist	A user who formulates scientific knowledge based on research. • Uploads genetic samples to GPAS, using the Electron Client. • Tracks local and global samples to determine if there are unique and concerning variants of pathogens.
Public Health Official	A user who administrates public health policy for a district or region, based on available data. • Generates reports and consumes dashboards to gain insight into significant changes in sample data warnings and prevent wider spread of infection. • Compares sample results from around the world to make informed public health policy decisions.

Pipeline components

GPAS supports various pipeline components for the analysis of viral pathogens and quality control outputs.

- [Viridian](#)
Viridian is an amplicon-aware genome assembly system designed for amplicon data.
- [Downstream analysis](#)
Once Viridian assembles the consensus genome, the results feed into various downstream analysis modules.
- [QC Outputs](#)
GPAS provides a suite of read-level QC outputs.

Viridian

Viridian is an amplicon-aware genome assembly system designed for amplicon data.

Viridian tags each read as belonging to a specific amplicon and discards low-quality amplicon reads. It then removes the primer at either end (20-30 bases, for ~300-400-base amplicons). These amplicons are reference-mapped, correcting the assembly for consistent deviations (genome polishing). By careful amplicon assembly, paying attention to primers, it can handle heterogeneity and also spot variance in the amplicon that may have otherwise been attributed to the primer.

Viridian workflow applies quality control metrics at the amplicon and base level, as follows:

- Viridian aligns reads to the reference (the default is MN908947.3). This removes reads that do not sufficiently match this sequence.
- Individual reads must be sufficiently long. For Illumina reads, `illumina_read_lengths` must be at least 50, by default.

- The start and end positions of reads (read-pairs) are used to infer which amplicon they belong to (for example, using nCoV-artic-v3.bed). Off-target reads (pairs) are discarded.
- Enough reads (pairs) must cover an amplicon. If not, all calls inside the region covered solely by that amplicon (not overlapping adjacent amplicons) are voided with Ns: min_depth, where the default depth is 50.
- Enough reads (pairs) must span at least 75% of the amplicon they belong to, excluding internal indels, where template length is the (alignment_end – alignment_start) / amplicon_length. This threshold is controlled by min_template_coverage_75, where the default is 80%. This filters out short fragments caused by PCR artefacts. If this threshold fails, the entire amplicon is discarded.
- After the assembly is constructed the original reads are mapped to it. Individual bases are voided with N if there is less than freq_threshold agreement. The default is 80%.
- After assembly, Viridian performs mild filtering: 70% of reads at a position need to support the consensus (FRS>0.7), and depth should be at least 10.

For more information on Viridian, see <https://github.com/iqbal-lab-org/viridian>

Downstream analysis

Once Viridian assembles the consensus genome, the results feed into various downstream analysis modules.

For more information on the Viridian system, see [Viridian](#).

GPAS supports the following modules for downstream analysis:

Table 1-2 Downstream analysis modules

Module	Description
Nextclade	Consensus sequences are pushed through Nextclade to provide Clade assignment (for example, 20a) and amino acid mutation calls. For more information on Nextclade, see https://clades.nextstrain.org/ GPAS reports Nextclade outputs, including: • Nextclade Clade: Assigned clade within Nextclade's system (for example, 20a). • QC score/status: Overall score assigned to the sequence, based upon Nextclade's individual QC rules. • Missing: Number of missing bases or "N's" within the consensus sequence. • Total substitutions: Total number of detected nucleotide substitutions. • Total deletions: Total number of detected nucleotide deletions. • Total insertions: Total number of detected nucleotide insertions. • Amino acid substitutions: Number of detected amino acid substitutions. • Amino acid deletions: Number of detected amino acid deletions.

Table 1-2 (Cont.) Downstream analysis modules

Module	Description
Pangolin	Pangolin provides lineage assignment, using USHER and pangoleARN (for example, B.1.1.7). Scorpio calls are also included in this module (https://github.com/cov-lineages/scorpio), which align with the WHO nomenclature (for example, Alpha, Delta, and so on). For more information on Pangolin, see https://cov-lineages.org/resources/pangolin.html GPAS reports the following Pangolin outputs: • Lineage: Most likely lineage assigned to a given sequence (for example, B.1.1.7). This is sensitive to missing data from key sites. • Conflict: Score that reflects the number of other categories to which the given sequence could be assigned. If the score is 0, then within that current decision tree, there is only one category it can be assigned. • Ambiguity score: Function of the quantity of missing data in a sequence. • Scorpio Call: Constellation assigned to the sequence by Scorpio. • Scorpio support: Proportion of defining variants which have the alternative allele in the sequence. • Scorpio conflict: Proportion of defining variants which have the reference allele in the sequence.
Aln2type	Aln2type assigns labels to samples, according to https://github.com/phe-genomics/variant_definitions, which are used by Public Health England to define and track variants of concern. For more information on Aln2type, see https://github.com/connor-lab/aln2type

QC Outputs

GPAS provides a suite of read-level QC outputs.

- **Mapped Reads:** Number of reads per sample.
- **Median Read depth:** Median number of reads covering reference sequence positions.
- **Reference Coverage @ Depth 10/20:** Percentage of reference sequence covered by at least 10/20 reads.
- **Dropped Amplicons:** Number of amplicons failing quality control.

For Illumina samples, GPAS also provides these read-level QC outputs:

- **Properly Paired reads:** Number of reads with expected pair orientation.
- **Median Insert size:** Median distance between the boundaries of paired reads covering the same fragment.
- **Reads within amplicons:** Number of properly-paired reads falling within expected coordinates of the same amplicon.

Get support

You can get support for GPAS in one of the following ways:

- Contact the **administrator who interacts with the GPAS expert.**

- Navigate to the **Oracle Support Portal** (<https://hsgbu.custhelp.com/>).
- In the GPAS User Portal, select the **Help** icon to access Oracle Help Center.

2

Get started

Once your organization signs an agreement with Oracle, it submits a list of users who need to use the Global Pathogen Analysis Service (GPAS) for sequencing analysis. This list also includes the tags each user role can use.

The administrator then adds the appropriate accounts and sets up tags for the user roles. For more information, see [Tags overview](#).

- [Get started with Global Pathogen Analysis Service](#)
After an administrator adds your account to GPAS, you receive an email with details on your account and how to activate it. This account uses Oracle Identity Cloud Service (IDCS) and two-factor authentication.
- [Activate your account](#)
When your organization adds you as a GPAS user, you receive an email that prompts you to activate your account.
- [Two-factor authentication](#)
The first time you sign in to GPAS, you must set up two-factor authentication. Two-factor authentication ensures the security of GPAS by requiring you to enter something you know (your password) and something you have (your one-time passcode).
- [Sign in to Global Pathogen Analysis Service](#)
You sign in to GPAS through your web browser.

Get started with Global Pathogen Analysis Service

After an administrator adds your account to GPAS, you receive an email with details on your account and how to activate it. This account uses Oracle Identity Cloud Service (IDCS) and two-factor authentication.

To get started with GPAS:

1. Activate your GPAS user account, as described in [Activate your account](#).
2. Set up two-factor authentication (2FA), as described in [Two-factor authentication](#).
3. Sign in to your user account, as described in [Sign in to Global Pathogen Analysis Service](#).

Activate your account

When your organization adds you as a GPAS user, you receive an email that prompts you to activate your account.

To activate your GPAS account:

1. In your email application, locate the Welcome message sent to you from Oracle.

 Note:

If you cannot find the Welcome email message, check your Junk folder in your email application. If you still cannot locate it, use the Oracle Support Portal to request a new message.

2. In the Welcome message, select **Activate Your Account**.

For security reasons, the link expires after 24 hours (your message contains exact details on the expiration range). If your link expires before you select it, contact support using the Oracle Support Portal.

3. In Oracle Identity Cloud Service (IDCS), enter a password and then confirm that password. When you are finished, select **Reset Password**.

A message displays, indicating your account has been activated. Sign in with your username and password. For more information, see [Sign in to Global Pathogen Analysis Service](#).

Two-factor authentication

The first time you sign in to GPAS, you must set up two-factor authentication. Two-factor authentication ensures the security of GPAS by requiring you to enter something you know (your password) and something you have (your one-time passcode).

A one-time passcode is a random code you receive that changes or expires within a set period of time to ensure no one can guess it.

You can access your one-time passcode using email or through an authenticator app on your mobile device, such as Oracle Mobile Authenticator.

For more information on setting up two-factor authentication, see the following topics:

- [Enable two-factor authentication with email](#)
When you first sign in to GPAS, you can enable two-factor authentication using the email address for your account.
- [Enable two-factor authentication with a mobile app](#)
When you first sign in to GPAS, you can enable two-factor authentication using a mobile device.

Enable two-factor authentication with email

When you first sign in to GPAS, you can enable two-factor authentication using the email address for your account.

For details on signing in to GPAS, see [Sign in to Global Pathogen Analysis Service](#).

To enable two-factor authentication with your email address:

1. Sign in to GPAS with your username and password.
2. In Oracle Identity Cloud Service (IDCS), select **Enable Secure Verification**.
3. Select **Email** to receive an email with a one-time passcode.
4. Enter the one-time passcode into the Code text field.
5. Select **Verify Email Address**.

You can now use your email account to access your passcode when you sign in again.

Enable two-factor authentication with a mobile app

When you first sign in to GPAS, you can enable two-factor authentication using a mobile device.

For details on signing in to GPAS, see [Sign in to Global Pathogen Analysis Service](#).

To enable two-factor authentication with your mobile device:

1. Sign in to GPAS with your username and password.
2. In Oracle Identity Cloud Service (IDCS), select **Enable Secure Verification**.
3. Select **Mobile App**.
4. On the app store for your device, download and install one of the following:
 - The **Oracle Mobile Authenticator app**.
 - A third-party authenticator app, such as Google Authenticator™ or Microsoft® Authenticator.

Note:

If you use a third-party authenticator app, make sure you select the **Offline Mode or Use Another Authenticator App** check box before scanning the QR code.

5. In the authenticator app on your mobile device, select the option to add an account.
6. When prompted, hold your mobile device up to the screen and scan the QR code.

Note:

Make sure the authenticator app on your mobile device has permission to access your camera.

7. Once you see a message on your screen that indicates you have successfully enrolled the device you specified, click **Done** to continue to GPAS.

You can now use your mobile authenticator app to access your passcode when you sign in again.

Sign in to Global Pathogen Analysis Service

You sign in to GPAS through your web browser.

For more information on user roles in GPAS, see [Global Pathogen Analysis Service Users](#).

To sign in to GPAS:

1. Open GPAS User Portal by navigating to the URL in your browser. (Your administrator provided you with the URL when you activated your account.)

The sign in page opens.

2. Sign in to Oracle Identity Cloud Service (IDCS) with your username and password.
3. In the application launchpad, select **User Portal** to access GPAS User Portal.



Note:

If this is your first time signing in, GPAS prompts you to enable two-factor authentication. For more information, see [Two-factor authentication](#).

- [Sign out of Global Pathogen Analysis Service](#)
GPAS automatically signs you out after an amount of time that your administrator specifies. Once that time period expires, you need to sign back in. However, you can sign out at any time.

Sign out of Global Pathogen Analysis Service

GPAS automatically signs you out after an amount of time that your administrator specifies. Once that time period expires, you need to sign back in. However, you can sign out at any time.

To sign out of GPAS, select the user profile icon and then select **Sign Out**.

3

Set up your User Profile

Your User Profile contains options that you can specify to optimize the way that you interact with information in Global Pathogen Analysis Service (GPAS).

- [Set Profile preferences](#)
You can set preferences in your User Profile for Report Date Format, Report Time Format, Language, and Time Zone. This ensures the accuracy and relevance of the information you receive.
- [Manage notifications](#)
You can manage the GPAS email notifications you receive (for example, when a user in your organization uploads new samples or an administrator adds a backup user for the samples you own).

Set Profile preferences

You can set preferences in your User Profile for Report Date Format, Report Time Format, Language, and Time Zone. This ensures the accuracy and relevance of the information you receive.

To set preferences in your user profile:

1. Select **Profile** from the profile icon on the top-right corner of the GPAS interface (the square icon with your username initials).
2. In the **Report Date Format** field, select a date format (for example, **DD-MON-YYYY**).
3. In the **Report Time Format** field, select a time format (for example, **HH24:MI:SS**).

Note:

The default report time format is HH:MI:SSPM. This format uses a 12-hour clock with AM and PM.

4. In the **Language** field, select one of the following options:
 - **English**
 - **French**
 - **Spanish**
5. In the **Time Zone** field, select your time zone (for example, **US Eastern Time (-05:00)**).
6. When you have finished setting your preferences, select **Save**.

Manage notifications

You can manage the GPAS email notifications you receive (for example, when a user in your organization uploads new samples or an administrator adds a backup user for the samples you own).

To manage notifications:

1. Select **Manage Notifications** from the profile icon on the top-right corner of the GPAS interface (the square icon with your username initials).
2. Select or clear the checkbox next to each notification type, depending on whether you want to receive it.
3. Select **Save**.

4

Upload sample data

You can upload sample data to the Global Pathogen Analysis Service (GPAS) to determine if your samples are locally or globally unique.

- [Requirements for uploading samples](#)
GPAS defines specific requirements for uploading sample data.
- [Tags overview](#)
In GPAS, tags control who can access or view sample data. Your organization defines the tags that you have the ability to assign.
- [Examples of tags](#)
This topic describes examples of some tags that you encounter when using GPAS.
- [Upload client overview](#)
As a requirement, you must use the GPAS upload client to upload data.
- [Download the upload client](#)
As a requirement for uploading sample data, you download the upload client from the Upload Client page in GPAS.
- [Prepare a CSV file for upload](#)
To upload samples, you prepare a CSV file with the fields that GPAS needs to read your sample data.
- [Download CSV templates](#)
GPAS provides templates, in CSV file format, that contain the required data fields when you upload data. You can download and use these templates from the Upload Client page.
- [Upload a sample](#)
After you prepare your CSV file, you can upload sample data to GPAS.

Requirements for uploading samples

GPAS defines specific requirements for uploading sample data.

To upload samples to GPAS, you must download and install the upload client. See the following related information:

- For details on how to access the upload client, see [Download the upload client](#).
- For details on preparing data in a CSV file to upload to GPAS, see [Prepare a CSV file for upload](#).
- For details on uploading a sample, see [Upload a sample](#).

GPAS accepts samples from **Illumina** or **Nanopore** platforms, with the following requirements:

- fastq.gz file format

 Note:

If you use .bam file format, the upload client runs [Samtools](#) to convert BAM to FASTQ. GPAS then realigns your reads from the beginning.

- Two files per sample for Illumina.
- For Nanopore, files can be either concatenated single fastq or multiplexed samples.
- You must sequence the samples using any one of the following:
 - Sequence-Independent, Single-Primer Amplification (SISPA)
 - Artic V3 primers
 - Artic V4.0 or V4.1 primers
 - Midnight primers
 - AmpliSeq primers
 - For any other primers, contact gpas-accounts@ndm.ox.ac.uk

Tags overview

In GPAS, tags control who can access or view sample data. Your organization defines the tags that you have the ability to assign.

You must manually add tags when you upload samples, and specify at least one tag per sample. Once the sample finishes uploading, you can also assign additional tags to it when viewing it on the Samples page. GPAS does not automatically add tags to samples.

- For details on assigning tags to samples during an upload, see [Upload a sample](#).
- For details on adding tags to a sample that you own or have already uploaded, see [Add tags to a sample](#).

 Note:

You cannot add tags to a sample that you do not own.

On the Samples page, when you view information for an individual sample, the tags for that sample appear in the Tags column of the table. For details on viewing sample information, see [Review sample details](#).

Lineage	Tags	Warning	Status
AY.4	ENA_Data	No	Released
AY.4	ENA_Data	No	Released
AY.4	ENA_Data	No	Released

Tags have the following effects:

- Tags can grant access to some (or all) users within your organization. Tags can also grant access to external organizations (for example, a tag that enables members of your consortium to see each other's data, or shares your samples with government agencies).
- Some organizations choose to join the community of GPAS sharing organizations. These organizations have special tags that contribute data to the GPAS shared data pool. Only users with sharing tags are able to see this data.

Your organization defines the tags it uses. It also determines what tags each user can use. As a result:

- You can apply any tag that you have permission to use to any samples you upload.
- You can only see samples for the tags that you have permission to view.

For examples of tags that occur in GPAS, see [Examples of tags](#).



Note:

The following considerations apply:

- As a user, you cannot create new tags. To request a new tag or change the list of tags you can use, contact the Oracle Support Portal.
- Once you add a tag to a sample, you cannot remove it.

Examples of tags

This topic describes examples of some tags that you encounter when using GPAS.

By default, most organizations have access to the tags in the following table:

Table 4-1 Tag examples

Tag	Description
<code>your_organization_name</code>	A sample that belongs to your organization.
<code>project_name</code>	A sample that belongs to a project within your organization.

Table 4-1 (Cont.) Tag examples

Tag	Description
repeat	A sample that a user previously ran in GPAS.
test	A test sample.

Upload client overview

As a requirement, you must use the GPAS upload client to upload data.

The upload client is a tool that anonymizes sample data before uploading it into GPAS. Using [read-it-and-keep](#), the upload client positively selects SARS-CoV-2 data from samples and discards non-SARS-CoV-2 data. This ensures that no personally identifiable information (PII) uploads to the GPAS cloud.

The upload client renames all files that users upload, and removes or replaces local file headers. File names then use GPAS Global Unique Identifiers (GUID), so GPAS cannot learn your local laboratory identifier. You keep the original files and file names on your computer, while the files that the upload client renamed upload to GPAS. The upload client also creates a CSV file on your computer, linking the local file name with the unique identifier so that you can link the data from your FASTQ or BAM files.

Periodically, GPAS provides updates to the upload client.

The following image shows an example of the Upload Client page.

Current Client						Download History	My Tags	Granted Tags
Platform	Version	Created	Filename	Release Notes	Download			
Mac x64	1.0.8	06-JUN-22	gpas-1.0.8-mac-x64-stage.zip	• MD5: 34726B44314437496E667A507A6546396A3839565A413D3D	Download			
Windows x64	1.0.8	06-JUN-22	gpas-1.0.8-win-x64-stage.zip	• MD5: 6955555754436832357A39702F5A4E6C3547554D51513D3D	Download			
Linux 16.04	1.0.8	06-JUN-22	gpas-1.0.8-linux-x64-16.04-stage.tar.gz	• MD5: 6445416A64417750316D643939366554384F377A4C513D3D	Download			
Linux 18.04	1.0.8	06-JUN-22	gpas-1.0.8-linux-x64-18.04-stage.tar.gz	• MD5: 4F726772626F7130365536733154574C5173516E35673D3D	Download			



Note:

You cannot upload samples using an outdated upload client. You see a message in GPAS each time a new version gets released, with a prompt to download it.

For more information on downloading the upload client, see [Download the upload client](#).

Download the upload client

As a requirement for uploading sample data, you download the upload client from the Upload Client page in GPAS.

To download and access the upload client:

1. In GPAS, select **Upload Client** from the main menu.
2. On the Clients tab, under Current Client, navigate to the latest zipped file for the upload client for your platform. Select the **Download** icon next to it.

GPAS supports the upload client for the following platforms and file types:

- **Windows 10:** .zip file
 - **MacOS:** .zip file
 - **Linux Ubuntu 20.04:** .gz file
3. In your operating system, extract the files to a directory you specify.
 4. Navigate to the folder where you extracted your zipped file and click **GPAS.exe**.

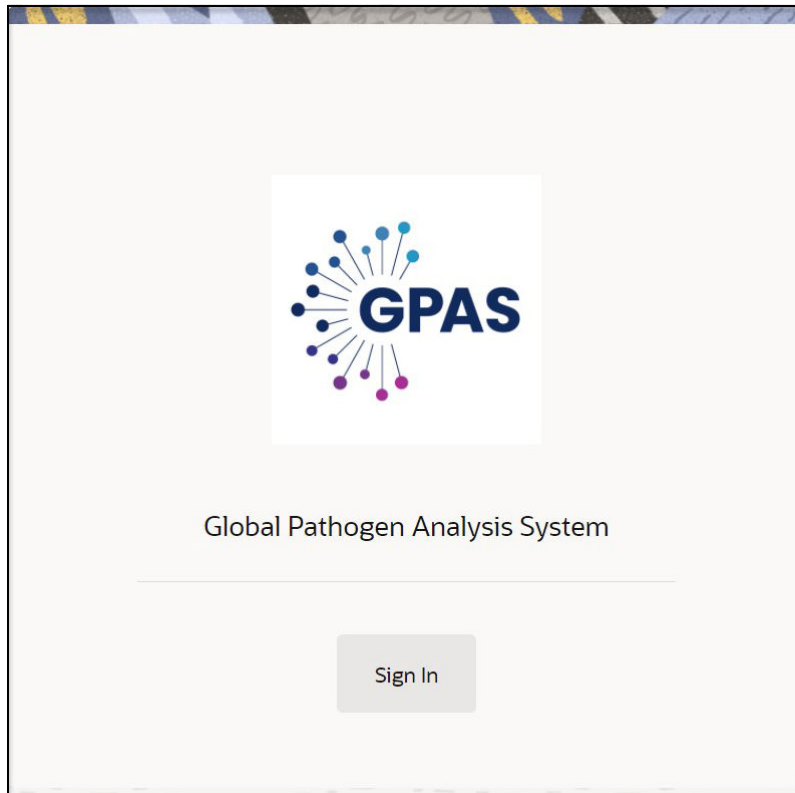
Note:

- If you use Linux, enter the following command in the directory where you extracted your .gz file:

```
$ ./gpas
```

- Make sure the GPAS-Uploader script and read-it-and-keep script have executable permission on your system.

The following page opens:



5. Select **Sign In**.
6. Sign in to Oracle Identity Cloud Service (IDCS) with your username and password.

 Note:

You sign in with the same username and password that you use to sign in to GPAS.

7. The first time you sign in to the upload client, you see a licensing agreement that prompts you to accept the terms. Select **I Accept** to continue.

The Upload Your Samples dialog box opens, where you can upload sample data to GPAS. For more information, see [Upload a sample](#).

Prepare a CSV file for upload

To upload samples, you prepare a CSV file with the fields that GPAS needs to read your sample data.

GPAS provides templates, in CSV file format, that contain the data fields GPAS requires when you upload data. For information on downloading these templates, see [Download CSV templates](#).

To prepare a CSV file for upload:

1. In the CSV file, enter the following fields and values:

- **batch:** Local batch number. A batch is any group of files uploaded together to GPAS. This field is required. For validation purposes, you must only use letters, numbers, underscores (`_`), periods (`.`), or dashes (`-`). This batch name is not shared with GPAS, but maps to a new GPAS batch GUID.
- **run_number:** Local sequencer run number. One GPAS upload batch may contain multiple sequencer runs. This field can be blank. However, if you enter a run number, for validation purposes, you must only use letters, numbers, underscores (`_`), periods (`.`), or dashes (`-`). This run number is not shared with GPAS, but maps to a new GPAS run GUID.

 Note:

When you upload data to GPAS, if you do not specify individual runs within a batch, GPAS treats all samples in a batch as one run.

- **sample_name:** Name of the sample. The sample name can be a meaningful name to your organization. This field is required. For validation purposes, you must only use letters, numbers, underscores (`_`), periods (`.`), or dashes (`-`). Each sample name must be unique. GPAS cannot upload the same sample name more than once from your organization. The sample name is not shared with GPAS, but maps to a new GPAS sample name GUID.
- **fastq:** File name of the FASTQ file, in the following format:
`GUID.fastq.gz`

There are separate CSV upload file templates for Illumina, Nanopore, and BAM file uploads. For more information, see [Download CSV templates](#).
 - For Nanopore samples, use a single **fastq** field.
 - For Illumina samples, use the fields **fastq1** and **fastq2** for the paired reads for each sample.
 - For BAM files, use the **bam** field heading, and enter the BAM file name.
- **control:** Control value for the sample. This is case sensitive. For validation purposes, this can only be **positive** (for positive control samples) or **negative** (for negative control samples). Otherwise, leave this field blank.
- **collection_date:** Date of collection for the samples. For validation purposes, this must be in ISO format (YYYY-MM-DD), no earlier than 2019-01-01, and no later than today's date.

 Note:

Some organizations do not have permission to share the day of sample collection with GPAS. These organizations have a date masking feature as part of their contract. You can provide the correct sample collection date in the upload metadata, and the upload client automatically masks the date to the beginning of the week, or the beginning of the month, before uploading to GPAS.

- **tags:** Tags you assign to a sample. As a requirement, you must specify at least one tag to each sample. For more information, see [Tags overview](#).

 Note:

Use a colon (:) to separate multiple tags.

- **country:** Country of origin for a sample. This field is required. For validation purposes, this must be the ISO 3166-1 country name in Alpha-3 format (for example, **USA** for the United States of America or **GBR** for Great Britain). You can find a list of the latest country codes online (for example, https://en.wikipedia.org/wiki/ISO_3166-1_alpha-3).
- **region:** Top-level subdivision of the country or origin for a sample. For validation purposes, this must be an ISO 3166-2 subdivision name (for example, Oxfordshire, New York, or Atacama). You can find a list of the latest subdivision codes online (for example, https://en.wikipedia.org/wiki/List_of_ISO_3166_country_codes).

 Note:

- This field is optional.
- Region data is only visible to users with access to the sample.
- Region data is not visible as part of the Shared Data Pool.

- **district:** More specific location information, such as a metropolitan area. This is free text. For validation purposes, you must enter an alphanumeric value, including letters, numbers, underscores (_), or colons (:). District data is only visible to users with access to the sample.

 Note:

This field is optional.

- **specimen_organism:** The pathogen (must be **SARS-CoV-2**).
- **host:** Host organism (must be **human**).
- **instrument_platform:** Must be either **Nanopore** or **Illumina**. This is case sensitive.

 Note:

You can only include one value for instrument_platform per CSV file because there are separate CSV upload templates for BAM, Illumina, and Nanopore.

- **primer_scheme:** Primer scheme. This must always be **auto**, and is case sensitive. GPAS auto-detects the primers in your sample (primer scheme is included in your GPAS results).

 Note:

Check that your fields contain the correct information and formats. For example, if a field is case sensitive, uses a specific date format or ISO code, or requires a Global Unique Identifier (GUID), this should match the requirements for upload. GPAS flags incorrect entries as errors.

You can run a test validation of your CSV file using the following URL: <https://github.com/GenomePathogenAnalysisService/gpas-uploader-validate>

2. Save your file.

The following CSV file shows an example of sample data ready to upload:

batch	run number	sample name	fastq1	fastq2	control	collection date	tags	country	region	district	specimen organism	host	instrument platform	primer scheme
B e Sef09	12	sample05_5f10	sample05_5f10_1.fastq.gz	sample05		2022-05-06	lab_tag	CHL	Atacama	Copapo	SARS_Cov_2	human	Illumina	auto
B e Sef09	12	sample01_36c3	sample01_36c3_1.fastq.gz	sample01	positive	2022-05-07	lab_tag:repeat	GBR			SARS_Cov_2	human	Illumina	auto
B e Sef09	12	sample03_5d8c	sample03_5d8c_1.fastq.gz	sample03	negative	2022-05-09	lab_tag	GBR	Oxfordshire		SARS_Cov_2	human	Illumina	auto
B e Sef09	12	sample04_632f	sample04_632f_1.fastq.gz	sample04		2022-05-05	lab_tag	GBR	Oxfordshire	Aylesbury	SARS_Cov_2	human	Illumina	auto
B e Sef09	13	sample07_5204	sample07_5204_1.fastq.gz	sample07		2022-05-03	test	GBR	Wiltshire	Trowbridge	SARS_Cov_2	human	Illumina	auto
B e Sef09	13	sample02_e0c0	sample02_e0c0_1.fastq.gz	sample02	positive	2022-05-07	lab_tag	USA	New York	Albany	SARS_Cov_2	human	Illumina	auto

Download CSV templates

GPAS provides templates, in CSV file format, that contain the required data fields when you upload data. You can download and use these templates from the Upload Client page.

To download CSV templates from GPAS:

1. On the Upload Client page, select **Get CSV Templates**.

This downloads a zip file to your machine.

2. In your operating system, navigate to the zip file and select the template you want to use.

The following templates are available:

- For BAM files: `client_upload_template_BAM_<upload_client_version>.csv`
- For Illumina: `client_upload_template_PE_<upload_client_version>.csv`
- For Nanopore: `client_upload_template_SE_<upload_client_version>.csv`

 Note:

When you upload your CSV file to GPAS, make sure it resides in the same directory as your BAM or FASTQ files. For more information, see [Prepare a CSV file for upload](#).

Upload a sample

After you prepare your CSV file, you can upload sample data to GPAS.

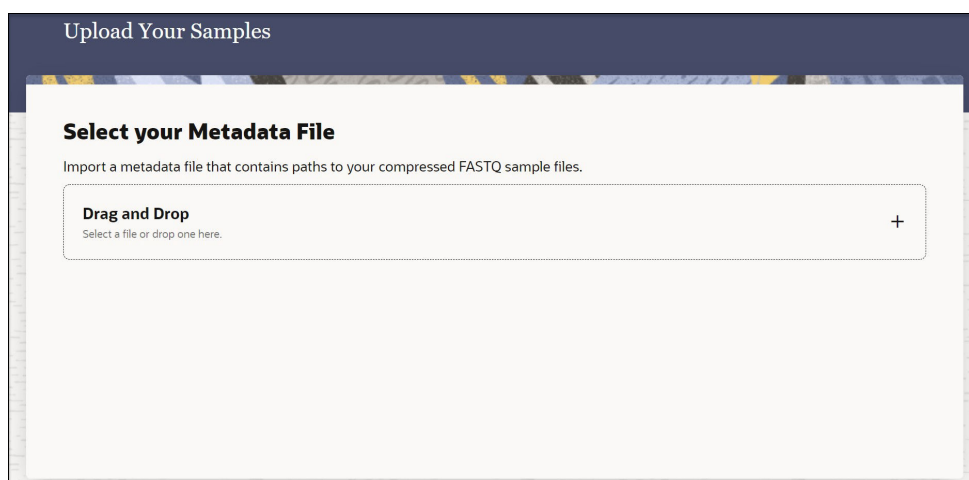
Note:

As prerequisites:

- You must download the upload client from the Upload Client page in GPAS. For more information, see [Download the upload client](#).
- You must prepare a CSV file with fields and values that GPAS can read. For more information, see [Prepare a CSV file for upload](#).

To upload a sample to GPAS:

1. Sign in to the upload client on your computer.
2. Drag and drop your CSV file into the field or browse to the file from your directory.



Note:

As a requirement, the FASTQ or BAM files for your samples must reside in the same directory as your CSV file, so GPAS can read them.

3. The upload client reads the CSV file and checks it for data integrity. It then uploads the FASTQ or BAM files that you associated with the CSV file. Note that depending on batch size, this can take some time (approximately one minute per sample).
4. Once the upload completes, you can view the samples you uploaded in the Samples page of GPAS. For more information, see [Review sample details](#).

5. The upload client places a `sample_names.csv` file in the same directory as your upload files and metadata. This file maps the GPAS GUID and the sample name you provided in your CSV file.

Every time you run a batch of samples in GPAS, the upload client provides a new mapping file. The file name is always `sample_names.csv`. It contains one line for each sample you uploaded in the batch, in the following format:

```
<sample_name>, <GPAS GUID>
```

The file contains the following fields:

- **local_batch:** The batch name that you specify in your CSV upload file. For example, B-e5ef09.
- **local_run_number:** The run number that you specify in your CSV upload file. For example, 12.
- **local_sample_name:** The sample name that you specify in your CSV upload file. For example, sample05_5f10.
- **gpas_batch:** The batch name that GPAS assigns. For example, A-41da1c8a.
- **gpas_run_number:** The run number that GPAS assigns. For example, f9178.
- **gpas_sample_name:** The GUID for the sample that GPAS assigns. For example, 571c01f1-b8be-9dfe-6cd3-f6e4568f28d5B-7d57c8.

5

Review sample details

You can review sample details to identify SARS-CoV-2 variants, mutations, or other warnings, so that you can formulate scientific knowledge.

- [Sample details overview](#)
Once you upload a sample, you can see specific information about that sample, such as its status, lineages, mutations, and other critical information.
- [Verify the run quality status of a batch](#)
After a batch that you upload to GPAS completes analysis, you must pass or fail the runs within the batch before releasing samples.
- [Export sample data as a report](#)
You can export sample data output as a report for offline analysis.
- [Transfer ownership to another user](#)
If you own a batch or sample in GPAS, you can transfer that ownership to another user. This gives them permission to perform actions on the batch or sample, such as changing sample status, adding tags, and transferring their ownership.
- [Change sample status](#)
If you own a batch or sample in GPAS, you can change the sample status to Released.
- [Add tags to a sample](#)
If you own a batch or sample in GPAS, you can add tags to the sample.

Sample details overview

Once you upload a sample, you can see specific information about that sample, such as its status, lineages, mutations, and other critical information.

The following image shows an example of the Samples page.

	Batch	Owned By	Uploaded 	Runs	Samples	Quality	Lineages	Status
☐	ENA-1dd5781	marc.brouard@ndm.ox.ac.uk	25-MAY-22		20	Unconfirmed (20)		Uploaded (20)
☐	ENA-70d7868	marc.brouard@ndm.ox.ac.uk	24-MAY-22		200	Unconfirmed (200)		Uploaded (200)
☐	ENA-beac102	marc.brouard@ndm.ox.ac.uk	11-MAY-22		200	Passed (200)		Uploaded (200)
☐	ENA-7978e2d	marc.brouard@ndm.ox.ac.uk	14-APR-22		1000	Passed (1000)		Uploaded (1000)
☐	ENA-e3e8816	marc.brouard@ndm.ox.ac.uk	14-APR-22		1000	Passed (1000)	AY.4 (471), AY.4.2 (79) , +52	Error (6) Released (994)
☐	ENA-04dcd30	marc.brouard@ndm.ox.ac.uk	14-APR-22		1000	Passed (1000)	AY.4 (349), BA.1 (313) , +51	Error (5) Released (997)
☐	ENA-e15f960	marc.brouard@ndm.ox.ac.uk	14-APR-22		1000	Passed (1000)		Uploaded (1000)
☐	ENA-8e06c08	marc.brouard@ndm.ox.ac.uk	14-APR-22		1000	Passed (1000)		Uploaded (1000)

- [Sample status descriptions](#)
You can view the status of samples, such as Uploaded, In Progress, Unreleased, Released, Error, or Abandoned, in GPAS.
- [Sample report descriptions](#)
You can view interactive reports for individual samples.
- [Sample view customization options](#)
You can perform a search for samples using any of the following options:

Sample status descriptions

You can view the status of samples, such as Uploaded, In Progress, Unreleased, Released, Error, or Abandoned, in GPAS.

When you upload sample data, GPAS treats each group of samples you upload together as a batch. On the Samples page, in the Batches tab, you can view the status of the batch, as a whole. Note that individual samples within a batch may have different statuses.

The following table describes the different statuses that GPAS assigns to samples.

Table 5-1 Sample status

Status	Description
Uploaded	You uploaded the sample to GPAS, but analysis is not yet complete.
In Progress	Analysis is in progress for the sample.
Unreleased	The pipeline analyzed the sample and you can now view the results. However, the pipeline did not release the sample to the GPAS Shared Data Pool.
 Note: In order to release a sample, you must apply sharing tags to it. When analysis is complete (the sample has a status of Unreleased), you manually release the sample within the GPAS portal. • For information on releasing samples, see Change sample status. • For more information on using tags in GPAS, see Tags overview.	
Released	The owner of the sample has released the sample to the GPAS Shared Data Pool.
Error	One or more samples in this batch did not correctly run for analysis through GPAS. This can happen for several reasons, including poor coverage within the sample or incompatible metadata during upload. If this happens and you do not know why, contact the support team.
Abandoned	GPAS cannot process the data for this sample.

Sample report descriptions

You can view interactive reports for individual samples.

To view details for a specific sample, navigate to the batch you want to see and select **View**. Within the batch, navigate to the sample and click **Report**.



Note:

Depending on the status of the sample, certain information may not yet be available. For example, if the status of a sample is Uploaded, and you select one of the other tabs (for example, Mutations), GPAS displays a message that indicates the data is still processing, and to check again later to view the results.

Table 5-2 Sample details

Report	Description
Details	Provides a summary of the metadata for the sample, as well as the results from the downstream analysis modules (for example, Pangolin or Nextclade). You can also download individual sections of the downstream analysis in JSON file format. For more information on downloading reports, see Export sample data as a report .
Lineage Map	A global map with a representation of the number of samples that match the lineage of your sample. Note that you can only access samples within GPAS if your permissions allow you to view them. Other users can assign these permissions to you through sharing tags. For more information, see Tags overview .
Relatedness	A bar chart that shows relatedness between the sample and other samples within GPAS, using Single Nucleotide Polymorphisms (SNP). Related samples display in the following groups: • Identical (No differences) • Highly Similar (1 SNP difference) • Very Similar (2 SNP differences) • Similar (3 SNP differences)
Mutations	A list of amino acid mutations for the sample. You can download this list for offline analysis. For more information on downloading reports, see Export sample data as a report .
Sample Reads	A bar chart that provides details about the amplicon set in the sample. The y-axis displays the number of reads (each bar chart column also represents the number of reads). The x-axis records the nucleotide positions across the genome. The bar chart color codes each column by gene and records the median and minimum read depths in the table. This view contains two panels: • The lower panel displays the entire amplicon set. You can scroll across sections of the amplicon set to view sections of the data. • The upper panel displays the section of the amplicon set you scrolled to in the lower panel.

Table 5-2 (Cont.) Sample details

Report	Description
Tags	The tags that the sample owner associates with a sample (for example, <code>pos_control</code> or <code>qc_fail</code>). Tags control who can access or share a sample. Only the user who owns the sample can add tags to it. You cannot remove tags once you assign them to a sample. For more information on tags, see Tags overview .
History	A log that provides a history of activity for the sample, such as when a sample status changes. Each action shows the Table Name, Field, Old Value, New Value, Action, and Time Stamp.

 Note:

You must have access to a sample to view this tab.

Sample view customization options

You can perform a search for samples using any of the following options:

Table 5-3 Search options

Component	Description
Select columns to search	Select the magnifying glass icon () to filter for the following parameters: • Batch • Runs • Samples • Quality • Lineages • Status • Owned By
Search Report text box	Type the search query for the filter you applied. Optionally, enter a generic search team for the samples you want to view. To execute your search, press Enter.

Verify the run quality status of a batch

After a batch that you upload to GPAS completes analysis, you must pass or fail the runs within the batch before releasing samples.

For more information on uploading files to GPAS, see [Upload sample data](#).



Note:

You must be the owner of a batch to pass or fail runs.

To verify the run quality status of a batch:

1. On the Samples page, in the Batches tab, do one of the following:
 - To pass or fail all the runs in a batch, navigate to the batch and select it.
 - To pass or fail individual runs in a batch, navigate to the batch, and in the Runs column, select the link showing the number of runs in the batch to open the Runs dialog box. Then, navigate to an individual run and select **View**.



Note:

When you upload data to GPAS, if you do not specify individual runs within a batch, GPAS treats all samples in a batch as one run.

2. In the Run view, select one of the following:
 - If the run passes quality control, select **Pass Run**.
 - If the run fails quality control, select **Fail Run**.
3. Select one of the following:
 - To cancel, select **Cancel**.
 - To confirm the pass or fail of the run, select **Confirm**.
 - To confirm the pass or fail of the run and proceed to the next run, select **Confirm and Go to Next Run**.



Note:

This option is not available for the final run in a batch.

When you finish verifying the run quality status, the Quality Status column on the Batches tab updates to show the number of runs within the batch that are Passed or Failed. Additionally, in the Run view, the date stamp at which you passed or failed the run displays on the top-right corner of the page.

You can now release your samples. For more information, see [Change sample status](#).



Note:

If you previously failed a run and want to update it to Passed, in the Run view, navigate to the top-right corner of the page. Next to the date stamp, select **Update**. Select **Pass Run** and then select **Confirm**. Your run is now Passed.

Export sample data as a report

You can export sample data output as a report for offline analysis.

- [Export batches](#)
You can export sample data as a report for a batch or multiple batches.
- [Export multiple samples](#)
You can export sample data as a report for multiple samples.
- [Export an individual sample](#)
You can export sample data as a report for an individual sample.
- [Export sections of sample data as JSON](#)
You can export sections of sample data for individual downstream analysis modules as JSON files, instead of the entire JSON report.

Export batches

You can export sample data as a report for a batch or multiple batches.

To download reports for batches:

1. On the Samples page, navigate to the Batches tab and do one of the following:
 - Select the check box next to each batch that you want to download.
 - Select the check box at the top of the list to multiselect all batches.
2. Select the **Download** drop-down menu and then select one of the following file options:
 - **CSV** (options include **Summary** and **Mutations - Compact**)
 - **TSV** (options include **Summary** and **Mutations - Compact**)
 - **JSON** (options include **Sample Expanded** and **Summary**)
 - **VCF**

Note:

A Summary report contains all the details for each sample in the batches you select. A Sample Expanded report contains all of the JSON from the SP3 pipeline analysis.

Note:

If you attempt to download a large amount of files, GPAS prepares your files as a background job. Once this completes, you receive a notification that your files are ready. You can then select **My Downloads** from the main menu to download your files.

Export multiple samples

You can export sample data as a report for multiple samples.

To download a report for multiple samples:

1. On the Samples page, navigate to the Batches tab and select a batch.
2. Do one of the following:
 - Select the check box next to each sample that you want to download.
 - Select the check box at the top of the list to multiselect all samples.
3. Select the **Download** drop-down menu and then select one of the following file options:
 - **PDF** (options include **Sample Summary**)
 - **CSV** (options include **Summary**, **Mutations - Long Form**, and **Mutations - Compact**)
 - **TSV** (options include **Summary**, **Mutations - Long Form**, and **Mutations - Compact**)
 - **JSON** (options include **Expanded Details** and **Summary**)
 - **BAM**
 - **FASTA** (options include **FASTA.gz** and **Combined FASTA**)
 - **VCF**

 Note:

A Summary report contains all the details for each sample you select. An Expanded Details report contains all of the JSON from the SP3 pipeline analysis.

 Note:

If you attempt to download a large amount of files, GPAS prepares your files as a background job. Once this completes, you receive a notification that your files are ready. You can then select **My Downloads** from the main menu to download your files.

Export an individual sample

You can export sample data as a report for an individual sample.

To download a report for an individual sample:

1. On the Samples page, navigate to the Batches tab and select the batch.
2. Navigate to the sample you want to download and select it.

The page for the sample opens, displaying details about the sample.

3. In the top-right corner, select the **Download** drop-down menu and then select one of the following file options: **Summary PDF**, **JSON**, **BAM**, **FASTA**, **VCF**, **Mutations - Long Form** (available options include **CSV** or **TSV**), or **Mutations - Compact** (available options include **CSV** or **TSV**).

 Note:

- You can also export individual sections of a sample report as JSON for a specific downstream analysis module. For more information, see [Export sections of sample data as JSON](#).
- You can only export reports for samples that completed analysis. A sample status of Unreleased or Released indicates completion. If you see a sample status of Uploaded or Error, the analysis did not complete, and you cannot download a JSON file. For more information on sample status, see [Sample details overview](#).

Export sections of sample data as JSON

You can export sections of sample data for individual downstream analysis modules as JSON files, instead of the entire JSON report.

To download sections of sample data as JSON files:

1. On the Samples page, navigate to the Batches tab and select a batch.
2. Navigate to the sample you want to view and select the sample.
The page for the sample opens, displaying details about the sample.
3. On the Details tab, scroll down to the bottom of the page and navigate to one of the sections (for example, **Pangolin Output**, **Nextclade Output**, **Aln2type Output**).
4. Select the download icon on the right-hand side to download the JSON file.
5. Optionally, you can expand the downstream analysis module section to view the contents of the JSON file before you download it. To do this, select the arrow icon on the left-hand side.



▼ Pangolin Output  Pangolin

▢ ambiguity_score - 0.8218475073313782
▢ conflict - 0
▢ lineage - BA.1
▢ note - scorprio call: Alt alleles 36; Ref alleles 0; Amb alleles 19; Oth alleles 4
▢ pangoleARN_version - 2022-02-28
▢ pango_version - v1.2.127
▢ pangolin_version - 3.1.20
▢ sampleName - ERR7941496
▢ scorprio_call - Probable Omicron (BA.1-like)
▢ scorprio_conflict - 0
▢ scorprio_support - 0.6102
▢ status - passed_qc
▢ taxon - ERR7941496
▢ version - PLEARN-v1.2.127

> Nextclade Output  Nextclade

> Aln2type Output  Aln2type

Transfer ownership to another user

If you own a batch or sample in GPAS, you can transfer that ownership to another user. This gives them permission to perform actions on the batch or sample, such as changing sample status, adding tags, and transferring their ownership.



Note:

You cannot transfer ownership of a batch or sample you do not own.

To transfer ownership of a sample:

1. On the Batches tab or the My Organisation tab in the Samples page, navigate to the batches or samples you want to transfer.
2. Do one of the following:
 - To select one sample at a time, select the check box next to each sample.
 - To select all the samples in the list, if you are viewing a batch of samples in the Batches tab, select the check box at the top of the column.
 - To select a single batch or multiple batches, in the Batches tab, select the check box next to each batch.
3. Select **More Actions** and then select **Transfer Batches** for a batch or **Transfer Samples** for a sample.



Note:

- If you select a batch or sample that you do not own, the More Actions button is unavailable.
- If you select multiple batches or samples and you do not own all of them, when you attempt to transfer ownership, a message indicates that you do not own all the samples you selected.

4. Select the **User** drop-down menu and then select a user.
5. Select **Save** to transfer ownership to that user.

Change sample status

If you own a batch or sample in GPAS, you can change the sample status to Released.

When you change a sample status to Released, you add that sample to the GPAS Shared Data Pool. You can then see genetic relatedness about this sample, as well as allow other members of the GPAS Sharing Community to see how your sample relates to other samples in the Shared Data Pool.



Note:

You cannot change the status of a sample you do not own.

To change the status of a sample:

1. Navigate to the samples you want to change from the Batches tab or the My Organisation tab (depending on your role).
2. Do one of the following:
 - To select one sample at a time, select the check box next to each sample.
 - To select all the samples in the list, if you are viewing a batch of samples in the Batches tab, select the check box at the top of the column.
 - To select a single batch or multiple batches, in the Batches tab, select the check box next to each batch.
3. Select **More Actions** and then select **Change Status**.



Note:

- You cannot change the status of samples you do not own.
- If you select a batch or sample that you do not own, the More Actions button is unavailable.
- If you select multiple batches or samples and you do not own all of them, when you attempt to change status, a message indicates that you do not own all the samples you selected.

4. To release a sample, select **Released**.



Note:

If you select Released, the system releases the sample and adds the date and time in the Released On field in the Change Samples Status dialog box.

5. Select **Save** to change the sample status.



Note:

Once you release a sample, you cannot unrelease it. However, your organization can revoke sharing status, if needed.

Add tags to a sample

If you own a batch or sample in GPAS, you can add tags to the sample.

For more information on tags, see [Tags overview](#).

Note:

- You cannot add tags to a sample you do not own.
- Once you add a tag, you cannot remove it, as this may affect access for other users.

To add tags to a sample:

1. On the Batches tab or the My Organisation tab of the Samples page, navigate to the sample to which you want to add tags.
2. Do one of the following:
 - To select one sample at a time, select the check box next to each sample.
 - To select all the samples in the list, if you are viewing a batch of samples in the Batches tab, select the check box at the top of the column.
 - To select a single batch or multiple batches, in the Batches tab, select the check box next to each batch.
3. Select **More Actions** and then select **Assign Tags**.

Note:

- You cannot add tags to samples you do not own.
- If you select a batch or sample that you do not own, the More Actions button is unavailable.
- If you select multiple batches or samples and you do not own all of them, when you attempt to add tags, a message indicates that you do not own all the samples you selected.

4. Select the **Tags** drop-down menu and then select a tag.
5. Select **Save** to add the tag.

6

Gain insights with dashboards

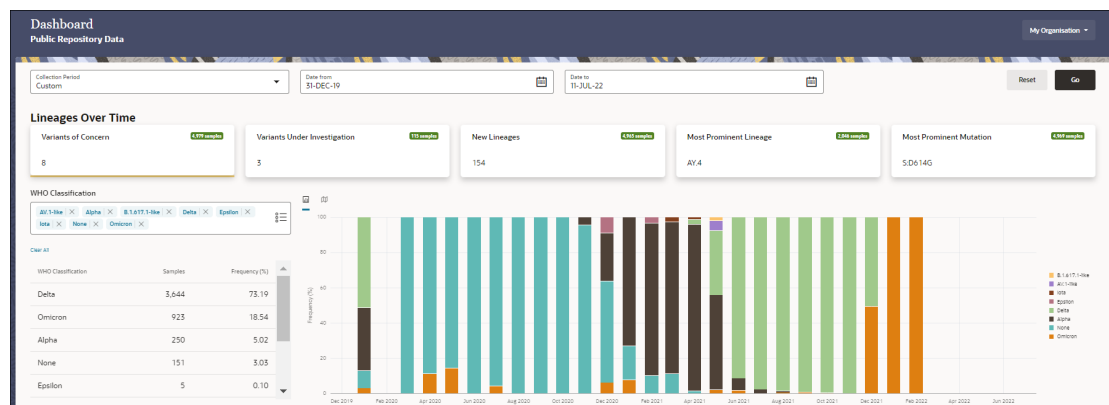
You can interact with dashboards to gain critical information such as collection date ranges, variants of concern, lineages, mutations, and geographic distributions. These analytics allow you to make informed public health policy decisions.

- [Dashboard overview](#)
As a Public Health Official, you use the dashboard to view interactive data for the samples that Research Scientists upload to GPAS.
- [Filter dashboard data](#)
When viewing a dashboard, you can filter the interactive data to show the information relevant to you.

Dashboard overview

As a Public Health Official, you use the dashboard to view interactive data for the samples that Research Scientists upload to GPAS.

The following image shows an example of the Dashboard page.



The Dashboard page shows an aggregate view of your sample data. By default, you can view the following dashboards:

- **My Organisation:** Displays dashboard analytics for samples in your organization that you can view.
- **Geographic Distribution:** Displays dashboard analytics for samples across the globe that you can view.

Dashboards contain category tiles that show dynamic summaries of sample data. When you select a tile, the summary chart displays for the category.



The following table describes the available categories:

Table 6-1 Dashboard categories

Category	Description
Variants of Concern	Shows variants with evidence of an increase in transmissibility, more severe disease (for example, increased hospitalizations or deaths), significant reduction in neutralization by antibodies generated during previous infection or vaccination, reduced effectiveness of treatments or vaccines, or diagnostic detection failures.
 Note: The UK Health Security Agency assigns this status according to the Variants of Concern list (https://www.gov.uk/government/collections/new-sars-cov-2-variant) using Aln2Type (https://github.com/connor-lab/aln2type).	
Variants Under Investigation	Shows variants with specific genetic markers that have been associated with changes to receptor binding, reduced neutralization by antibodies generated against previous infection or vaccination, reduced efficacy of treatments, potential diagnostic impact, or predicted increase in transmissibility or disease severity.
 Note: The UK Health Security Agency assigns this status according to the Variants of Concern list (https://www.gov.uk/government/collections/new-sars-cov-2-variant) using Aln2Type (https://github.com/connor-lab/aln2type).	
New Lineages	Shows lineages that appear for the first time in the GPAS data that you can view, for the date range you select at the top of the page. A lineage is a group of closely-related viruses with a common ancestor. SARS-CoV-2 has many lineages, and all of them cause COVID-19.

Table 6-1 (Cont.) Dashboard categories

Category	Description
Most Prominent Lineage	Shows the most prominent lineages from the sample data and the frequencies of their occurrence.
Most Prominent Mutation	Shows mutations over time, including the most prominent mutation from the sample data.

Filter dashboard data

When viewing a dashboard, you can filter the interactive data to show the information relevant to you.

To filter dashboard data:

1. On the top-right corner, click the drop-down menu and select one of the following dashboards:
 - **My Organisation**
 - **Geographic Distribution**
2. Select the **Collection Period** drop-down menu and select a collection period.
The Date from and Date to calendar control date fields automatically update to display the date range.
3. If you want to view a custom time period, click the **Date from** and **Date to** calendar controls and select your dates.
4. Select one of the following category tiles:
 - **Variants of Concern**
 - **Variants Under Investigation**
 - **New Lineages**
 - **Most Prominent Lineage**
 - **Most Prominent Mutation**

 Note:

Depending on whether you are viewing the My Organisation or Geographic Distribution dashboard, some category tiles, such as Variant of Concern, Variants Under Investigation, and New Lineages, may not be available.

5. Select the icon next to the WHO Classification field to open the Search dialog box, where you can select SARS-CoV-2 variants as search parameters to apply to your dashboard.
6. At the top of the dashboard, select **Go** to execute your filter.
7. To reset your dashboard, select **Reset**.

7

Track requests

You can submit requests in Global Pathogen Analysis Service (GPAS) to access samples or upload them to a public repository. Organization coordinators can evaluate sample access requests that users submit to determine who can access sample data.

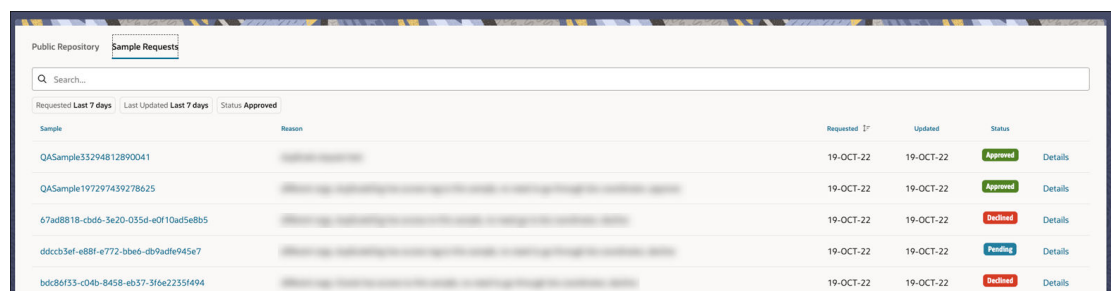
- [Requests in Global Pathogen Analysis Service](#)
You can request access to a sample that you do not own and cannot access through tags. You can also request to upload a sample to a public repository.
- [Request access to a sample](#)
You can request access to a sample that you do not own and cannot access through tags. These can be samples that your organization or another organization owns.
- [Request upload to a public repository](#)
Once you release a sample, you can use GPAS to make a request to upload it to a public repository.
- [Evaluate a sample access request](#)
As an organization coordinator, you can approve or decline requests to access samples from users in your organization or other organizations.

Requests in Global Pathogen Analysis Service

You can request access to a sample that you do not own and cannot access through tags. You can also request to upload a sample to a public repository.

When you make a request in GPAS, you can track its status in the My Requests page. This page includes a table with a list of all your requests, their latest status, and a link to view details.

The following image shows an example of the My Requests page.



Sample	Reason	Requested	Updated	Status	Details
QASample33294812890041	Requesting access to a sample that you do not own and cannot access through tags.	19-OCT-22	19-OCT-22	Approved	Details
QASample197297439278625	Requesting access to a sample that you do not own and cannot access through tags.	19-OCT-22	19-OCT-22	Approved	Details
67ad8818-cbd6-3e20-035d-e0f10ad5e8b5	Requesting access to a sample that you do not own and cannot access through tags.	19-OCT-22	19-OCT-22	Declined	Details
ddcb3ef-e88f-e772-bbe6-d9ad9e945e7	Requesting access to a sample that you do not own and cannot access through tags.	19-OCT-22	19-OCT-22	Pending	Details
bdc86f33-c04b-8458-e037-3f6e22334494	Requesting access to a sample that you do not own and cannot access through tags.	19-OCT-22	19-OCT-22	Declined	Details

By default, you can view the following tabs:

- **Public Repository:** Displays your requests to upload samples to a public repository. These requests progress to the public repository coordinator for approval. Each request has a status of Submitted, Registering, or Registered, depending on the stage of review. For more information on this request type, see [Request upload to a public repository](#).

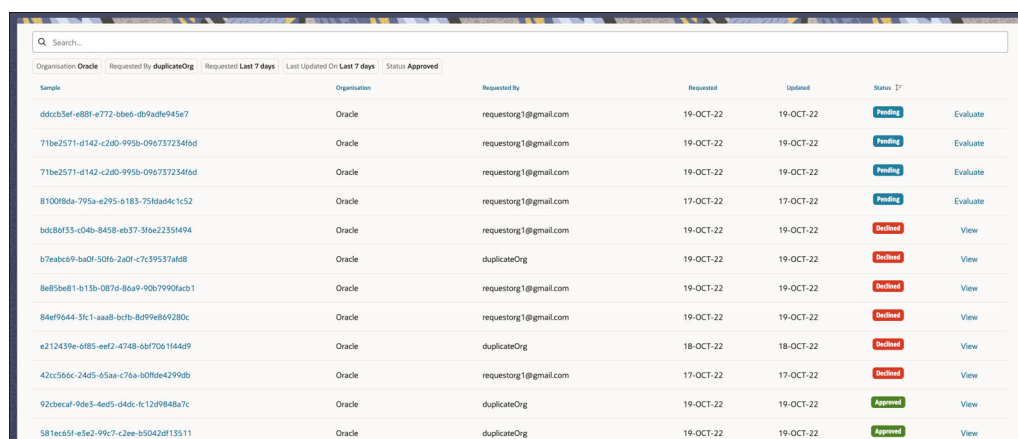
 Note:

Approval for uploading a sample to a public repository occurs outside of GPAS.

- **Sample Requests:** Displays your requests to access samples that you do not own and cannot access through tags. These requests progress to the coordinator for the GPAS organization that owns the sample for approval. Each request has a status of Pending, Approved, or Declined, depending on the stage of review. For more information on this request type, see [Request access to a sample](#).

If you are an organization coordinator, you receive sample access requests from users in your organization or other organizations. You evaluate these requests in the My Approvals page, where you can see the sample name, the organization that owns the sample, the requesting entity, request date, updated date, status, and details for the request. For more information on evaluating requests, see [Evaluate a sample access request](#).

The following image shows an example of the My Approvals page.



Sample	Organization	Requested By	Requested	Updated	Status	
ddcb3ef-e88f-e772-bbe6-d9ad9e45e7	Oracle	requestorg1@gmail.com	19-OCT-22	19-OCT-22	Pending	Evaluate
71be2571-d142-c2d0-995b-096737234f6d	Oracle	requestorg1@gmail.com	19-OCT-22	19-OCT-22	Pending	Evaluate
71be2571-d142-c2d0-995b-096737234f6d	Oracle	requestorg1@gmail.com	19-OCT-22	19-OCT-22	Pending	Evaluate
81008da-795a-e295-6183-75dad4c1c52	Oracle	requestorg1@gmail.com	17-OCT-22	17-OCT-22	Pending	Evaluate
bdc86f53-c04b-8458-eb37-36a2235494	Oracle	requestorg1@gmail.com	19-OCT-22	19-OCT-22	Declined	View
b7eabc69-ba0f-50f6-2a0f-c7c39537af88	Oracle	duplicateOrg	19-OCT-22	19-OCT-22	Declined	View
8e80be81-b13b-087d-86a9-90b7990facb1	Oracle	requestorg1@gmail.com	19-OCT-22	19-OCT-22	Declined	View
84ef9644-31c1-aaa8-8c7b-8d99e89280c	Oracle	requestorg1@gmail.com	19-OCT-22	19-OCT-22	Declined	View
e212439e-df85-ee12-4748-6b7061644d9	Oracle	duplicateOrg	18-OCT-22	18-OCT-22	Declined	View
42cc566c-24d5-d5aa-c76a-b0fde4299db	Oracle	requestorg1@gmail.com	17-OCT-22	17-OCT-22	Declined	View
92cbeccaf-9de3-4ed5-d4dc-fc1d9f98487c	Oracle	duplicateOrg	19-OCT-22	19-OCT-22	Approved	View
581ec65f-e5e2-99c7-c2ee-b5042df13511	Oracle	duplicateOrg	19-OCT-22	19-OCT-22	Approved	View

 Note:

You must be an organization coordinator to view the My Approvals page. The administrator for your organization assigns this role. For more information, see the *Manage organizations* topic in the *Global Pathogen Analysis Service Administrator Portal Guide*.

Request access to a sample

You can request access to a sample that you do not own and cannot access through tags. These can be samples that your organization or another organization owns.

To request access to a sample:

1. On the Relatedness tab of the Samples page, navigate to the sample you want to request and select it.

2. On the top-right corner of the Sample view page, select **Request Access**.
3. In the **Reason for Request** text box, type a justification for your request.
4. Select **Request Access**.



Note:

You can track the status of your request on the My Requests page by selecting **My Requests** from the main menu and then selecting the **Sample Access** tab. For more information, see [Requests in Global Pathogen Analysis Service](#).

Request upload to a public repository

Once you release a sample, you can use GPAS to make a request to upload it to a public repository.

To request uploading a sample to a public repository:

1. On the Batches tab or the My Organisation tab of the Samples page, navigate to the sample you want to upload to a public repository.
2. Do one of the following:
 - To select one sample at a time, select the check box next to each sample.
 - To select all the samples in the list, if you are viewing a batch of samples in the Batches tab, select the check box at the top of the column.
 - To select a single batch or multiple batches, select the check box next to each batch in the Batches tab.
3. Select **More Actions** and then select **Public Repository Upload**.



Note:

- You cannot upload samples you do not own to a public repository.
- If you select a batch or sample that you do not own, the More Actions button is unavailable.
- If you select multiple batches or samples and you do not own all of them, you can only upload the samples you own.
- If your samples are ineligible for upload to a public repository, the Ineligible Samples table displays the sample GUID and the reason for ineligibility (for example, you cannot upload control samples or samples that do not have a status of Released)

4. Select the **Public Repository** drop-down menu and then select **ENA**.
5. Select **Request Upload**.



Note:

You can track the status of your request on the My Requests page by selecting **My Requests** from the main menu and then selecting the **Public Repository** tab. For more information, see [Requests in Global Pathogen Analysis Service](#).

Evaluate a sample access request

As an organization coordinator, you can approve or decline requests to access samples from users in your organization or other organizations.



Note:

You must be an organization coordinator to view the My Approvals page and perform the steps in this task. The administrator for your organization assigns this role. For more information, see the *Manage organizations* topic in the *Global Pathogen Analysis Service Administrator Portal Guide*.

To evaluate a sample access request:

1. On the My Approvals page, navigate to a request with a status of Pending and select **Evaluate**.
2. In the Evaluate Request modal window, review the following fields to determine whether to grant the user or organization access to the sample:
 - **Reason For Request:** A justification for sample access from the requesting entity. This is free text.
 - **Sample Information:** Includes the date values for Collected, Uploaded, Shared to FN4, and Incognito End Date, Related Samples count, Access Tags, and a table that displays pipeline components, including Classifier, Result, and Version.



Note:

At the top of the modal window, you see an information flag if one or more of the following conditions apply:

- The request grants another organization access to the sample or access tag.
- You are approving sample access for a user in your organization, but another organization owns the sample and the coordinator for that organization must give final approval.

3. Select **Approve** to approve access to the sample, or **Decline** to deny access.